

MAINS MASTER PROGRAM (MMP) 2024

ENVIRONMENT-2

WATER - SECURITY, POLLUTION, MARINE ECOSYSTEM

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2. SYLLABUS

Conservation, environmental pollution and degradation, environmental impact assessment

3. PYQS OF WATER RELATED TOPICS

1. **[2023]:** Why is the world today confronted with a crisis of availability and access to freshwater resources? **[GS1, 10 marks, 150 words]**
2. **[2022]:** Explain the causes and effects of coastal erosion in India. What are the available coastal management techniques for combating these hazards **[GS3, 15 marks, 250 words]**
3. **[2020]:** What are the salient features of the Jal Shakti Abhiyan launched by the Government of India for water conservation and water security? **[GS3, 10 marks, 150 words]**
4. **[2019]:** Discuss the causes of depletion of mangroves and explain their importance in maintaining coastal ecology **[GS1, 10 marks, 150 words]**
5. **[2019]:** What is water stress? How and why does it differ regionally in India **[GS1, 15 marks, 250 words]**
6. **[2018]:** What are the consequences of spreading of 'Dead Zones' on marine ecosystem? **[GS1, 10 marks, 150 words]**
7. **[2018]:** The idea solution of depleting ground water resources in India is water harvesting system. How can it be made effective in urban areas. **[GS1, 15 marks, 250 words]**
8. **[2017]:** Not many years ago, river linking was a concept, but it is becoming a reality in the country. Discuss the advantages of river linking and its possible impact on the environment **[GS3, 10 marks, 150 words]**
9. **[2015]:** India is well endowed with freshwater resources. Critically examine why it still suffers from water scarcity. **[GS1, 12.5 marks, 200 words]**
10. **[2015]:** Discuss the Namami Gange and National Mission for Clean Ganga (NMCG) Programmes and causes of mixed results from the previous schemes. What quantum leaps can help the river Ganga better than incremental inputs? **[GS3, 12.5 marks, 200 words]**
11. **[2013]:** Enumerate the National Water Policy of India. Taking River Ganges as an example, discuss the strategies which may be adopted for river water pollution control and management. What are the legal provisions of management and handling of hazardous wastes in India? **[GS3, 10 marks, 200 words]**

4. WATER SECURITY

- Situation in India

- » India has **only 4% of the global freshwater resources** but it has to quench the thirst of **18% of the world's population**.
 - » **Water availability per capita has been falling at an alarming rate.**
 - It has come down from 5100 meter³ per capita in 1950 to 1,350 meter³ per capita currently.
 - It is further expected to go below 1,000 cubic meters in coming years.
 - » **Demand for water in India is largely driven by agriculture** and is expected to remain so in the coming future as well.
- Water demand by source (%)

Year	Irrigation (%)	Drinking (%)	Industry (%)	Energy (%)	Other (%)
2010	85	10	2	1	2
2025	82	10	3	3	2
2050	75	10	5	10	2

Source: Water-Nectar of Life, EnviStats - India 2018(Supplement on Environmental Accounts)
- » **Of the total gross cropped area, roughly half is irrigated and 63% of irrigated area depend on ground water** extraction for irrigation.
 - India is the **largest extractor** of the ground water in the world and this has resulted into ground water falling by average 0.4 meters every year.
 - » Thus, India is slowly moving towards **water stress** situation i.e. a situation when demand for water exceeds the available amount or when available amount is non-usable because poor quality restricts its use.

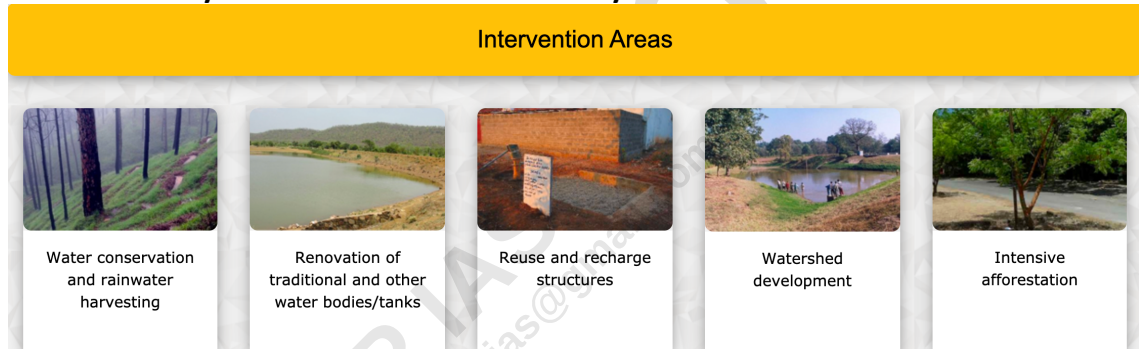
- Key factors behind the increasing Water Crisis:

- » **Increasing Population**-> decreasing per capita availability.
- » **Climate Change** -> **Deficient Rainfall** or extreme events.
- » **Geographical, Monthly and Yearly Variation in Rainfall**
- » **Inability to properly utilize water:**
 - India receives **4,000 BCM** (billion cubic meters) of annual precipitation, but the holding capacity of around 5,400 reservoirs across the country is **less than 300 BCM**
 - **Inefficient water utilization in Agriculture:** 80% of the water is used in agriculture, but in case of surface irrigation, 70% water is wasted.
 - **Urban India's inefficiency in water use** arises from inadequate old and dilapidated distribution networks, inefficient operations, inadequate metering, incomplete billing and collection, and a general state of poor governance. Further, not treating of waste water and under-pricing of urban water also contributes to wasteful use.
- » **Government Policies** (subsidy/free electricity; MSP for water guzzling crops) have exacerbated the water shortage by promoting wastage.
- » **Loss of water bodies like lakes, aquifers and catchment areas** to over exploitation, pollution, lack of conservation and encroachment.
- » **Environmental Pollution** has made many stretch of rivers unsuitable for drinking or agriculture purposes.

- **Negative Impact on**
 - » **Water Security**
 - » **Food Security**
 - » **Increasing Conflict between States** (e.g. Cauvery water dispute; Haryana-Delhi issue)
 - » **Social Unrest** also results from water crisis. During water scarcity the violence of water troubles also increase in the country.
 - » **Hampers Economic Growth**
 - » **Environmental Degradation and Biodiversity Loss**
 - **Heavy-metal Poisoning:** Extraction of water from deeper levels increase the chance of arsenic/fluoride poisoning.
 - Drought situation affects animals and plant life more than humans.

- **Steps Being Taken**

- » **National Water Mission** as part of NAPCC: Create a database; promote state & Citizen action; focus on vulnerable areas; improve water use efficiency; promote basin level integrated water resource management.
- » **Constitution of a new Jal Shakti Ministry** -> to take an integrated approach to the water problem
- » **Jal Shakti Abhiyan for water conservation. Key intervention areas** include:



- » **States are taking a number of initiatives**
 - **Rajasthan** has Mukhyamantri Jal Swavalamban Yojana
 - **Maharashtra** has initiated steps like **Jal Yukta Shivr.**
 - **Mission Kakatiya** in Telangana
- » **PMKSY** also has initiated several steps to **promote water use efficiency** throughout the country with an objective of '**more crop per drop**'.
- » **Atal Bhujal Yojna (ATAL JAL):** It is a world bank funded, central sector scheme aimed at **improving ground water management**. It is aimed at strengthening ground water governance mechanism in the participating states (initially 7 states in first five years) and various measures for ensuring the long-term sustainability of ground water resources.
- » **MGNREGA** has also played a key role in water conservation by creation of ponds, lakes and water conservation and storage mechanisms.
- » **Composite Water Management Index** by NITI Aayog to promote competitive and cooperative federalism among states in the field of water management.

- **Way Forward**

- » **Policy Level Changes**
 - **Bring an act on Ground Water management**

- **Proper audit of water** is the need of the hour to understand availability and requirement in each region of the country.
 - » Currently, proper planning is difficult due to the lack of latest data.
- **Need of Water Governance**
 - » Metropolitan cities should now set up urban water planning and management boards, a permanent body similar to urban development authorities, that regulate the supply, demand and maintenance of water services and structure.
- » **Supply Side Management - Properly Implement the Jal Shakti Mission**
 - **Source Augmentation:** Developing more micro dams to store water throughout the country.
 - **Deepening water bodies like lakes, ponds etc** for greater water storage and higher percolation.
 - Promote **watershed development** in water deficient region.
 - **Rainwater harvesting** has to be made an integral part of planning and development of new structure.
 - » It has been made compulsory for every building in Delhi in 2001 and in Chennai in 2003. Yet, even today important government buildings like the Supreme Court or NITI Aayog lack such facilities. Similarly, most government buildings in Chennai don't have a functioning RWH structure.
 - **More R&D focus** towards making desalination plants more economical and easily available.
 - **Waste-Water Recycling** is still to caught imagination of either authorities or the public in a big way.
 - » In India, it is estimated that 85 liters of water goes waste for every 100 liter utilized.
 - » For this to be successful, installed capacity of Sewage Treatment Plants (STP) has to double from the current situation. Presently, only 40.5% of sewage generated is treated. This treated water can be used for non-consumption purposes like gardening, flushing of toilets etc.
- » **Demand Size Management**
 - **Reforming Agri Sector** since agriculture consumes 80% of water used in the country
 - **Reducing Electricity Subsidy and Subsidy for water guzzling crops**
 - » Reducing these subsidies is tricky. But innovative steps can be taken - for e.g., the Punjab government is offering cash transfer to farmers for every unit of electricity they save to wean them away from pumping more water.
 - **Promoting Efficient Water Utilization**
 - » Making the efforts towards 'more crop per drop' more intensive. Investing more in promotion of micro-irrigation.
 - According to NITI Aayog, Micro-irrigation can reduce the usage of ground water by upto 20%.
 - » **A vigorous program on water efficiency standards**
 - A star rating system for water using appliance

- » Charge money for water -> though a sensitive issue, but can contribute heavily to efficient utilization
- **Changing cropping pattern** (rice in Punjab and Haryana, Sugarcane in Maharashtra) -> Price Policy -> Procurement issues etc.
 - » Water guzzling crops need to shift to geographically sustainable areas.
- **Technology** can be crucial in enhancing water use efficiency. Biotechnology can contribute towards development of less water guzzling varieties of crops.
- » **Design development around water management** i.e. make water management a central focus of a great deal of planning effort in agriculture, in urban development and many other areas.
 - **Incentive based water conservation efforts** - provide benefits for maintaining of water bodies, improving ground water level, conservation etc.
- » **Increased cooperation among different stakeholders** can promote coordinated efforts towards water conservation.
 - Among the users of aquifers
 - Among states at River Basin Level
 - Among all the users at village level
- **Conclusion**
 - » The solution to India's water crisis would come from **holistic management of both demand and supply side**. All stakeholders - Union government, state governments, NGOs, and citizens have to contribute to deal with the situation.

5. GROUND WATER AND RELATED ISSUES

Introduction: Global Situation

- » As per World Water Development Report, 2022, Ground water accounts for around 99% of the liquid freshwater on earth. It has continued to serve humankind for many millennia and currently around 50% of water used in domestic purpose and 25% of water used for irrigation globally comes from groundwater.
- » Yet, despite its enormous importance, this natural resource is often poorly understood, and consequently undervalued, mismanaged and even abused.

India's Situation:

Annual extractable groundwater availability in India (2017) is **393 BCM**.

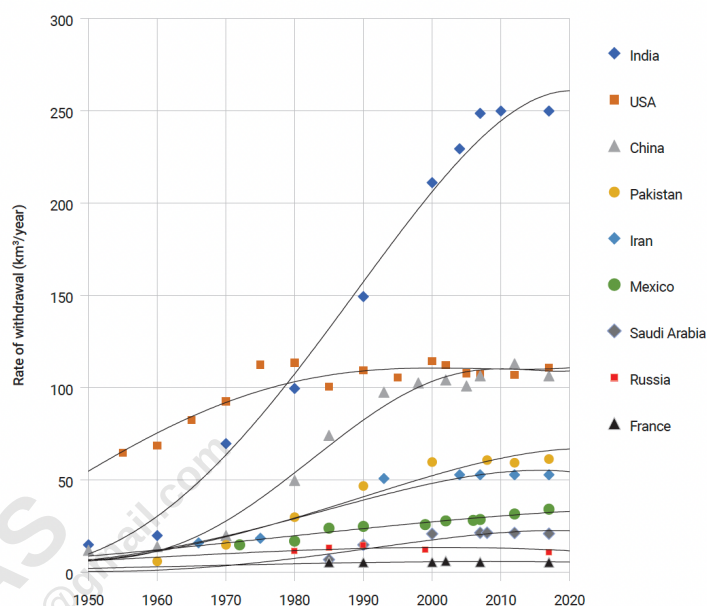
- » India is the largest user of ground water in the world, extracting **253 BCM** per year, which is 25% of the global ground water extraction. It extracts more groundwater than USA and China combined together.

Most of the ground water extracted in India is for **Irrigation** (228 billion Cubic Meter (BCM)) which accounts for **90%** of the total extraction.

- » In India, 60% of irrigation requirement is fulfilled by groundwater.

The remaining **10%** (225 BCM) is for drinking, domestic as well as industrial uses.

- » **Industrial use** accounts for **only 5%** of the total extraction



EVOLUTION OF GROUNDWATER EXTRACTION IN VARIOUS COUNTRIES

- **Satellite Gravimetry** has provided convincing evidence in support of the alarming rates of groundwater depletion.
- The data is supported by local level water table measurements in wells, where in 61% decline has been seen by CGWB.
- As per the 2022 assessment by the CGWB, 14% of assessments units in the country (1006/7089) have been categorized as 'Over-exploited' where the annual groundwater extraction is more than annual available Ground Water Resource. 4 States/Uts viz. Haryana, Punjab, Rajasthan, Dadra & Nagar Haveli and Daman & Diu have stage of Ground Water Extraction greater than 100%.

Key Challenges:

Depletion due to Over-extraction:

- Over the years, groundwater has become the dominant source of irrigation as well as for domestic purpose. T
- Installation of tube-wells have increased in north-western plains. Since the 1980s, 77% of the total addition to irrigation has come from tubewells.

- **Electricity Subsidy for agriculture and increased rural electrification** has also been a factor behind over-exploitation of ground water.
 - **Expansion of solar powered irrigation systems** which have led to very affordable cost of ground water extraction.
 - **Weak law and regulations** to prevent or limit diffuse groundwater pollution.
 - Industry that withdraws groundwater include manufacturing, mining, oil, and gas, power generation, engineering, and construction.
 - Bottled water industry is emerging as a major extractor.
 - **Destruction of wetlands, aquifers etc.** which used to act as water sinks and contributed to ground water recharge.
 - **Pollution:** Many sources of groundwater pollution are located in or near surface, but several other sources inject pollutants into the subsurface at greater depth below the surface.
 - Pollution due to Agriculture is widespread and difficult to control as it is a diffused source and includes large quantities of nitrates, pesticides, and other agronomics.
 - Industrial effluent discharge; Leaching through coal ash waste dumps; Fracking of natural gas, particularly in shallow aquifers can present considerable risk of groundwater contamination
 - **Irreversibility:** Once polluted, the aquifers tend to remain with polluted water.
 - **Climate Change:** CC impacts groundwater through impacting precipitation, leakage from surface water, sea water intrusion into coastal aquifers
- **Key Efforts for Groundwater:**
- **Recent Schemes:**
 - **Jal Shakti Abhiyan:** First launched in the year 2019, it focuses primarily upon effectively harvesting the monsoon rainfall through creation of artificial recharge structures, watershed management, intensive afforestation, awareness generation etc. JSA for the year 2023 was launched on 4th March 2023 with the theme "Source Sustainability for Drinking Water".
 - **Amrit Sarovar Mission** - launched in April 2022 - focuses on developing and rejuvenating 75 water bodies in each district of the country as part of celebration of Azadi ka Amrit Mahotsava.
 - **Atal Bhujal Yojana** is being implemented by central government in collaboration with states. It is being implemented in certain water stressed areas of Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh. The Primary aim of the scheme is **demand side management** through scientific means based on **water budgeting** of the area involving local communities at village levels leading to sustainable groundwater management in targeted areas.
 - **Institutions:**
 - **Central Ground Water Authority (CGWA)** has been constituted under Section 3(3) of the "Environment (Protection) Act, 1986" for the purpose of regulating and control of ground water by industries, mining projects, infrastructure, projects etc. in the country.
 - **CGWA** is also implementing **National Aquifer Mapping Program (NAQUIM)** in the country. These reports along with management plans are shared with States/Uts for suitable intervention.
 - **MoHUA** has formulated **Model Building By Laws (MBBL), 2016** for the states/ Uts, wherein adequate focus has been given on requirement of rainwater harvesting and water conservation measures. 35 states/Uts have adopted the features of the Model Bye Laws.
 - **Major and Medium projects** under Accelerated Irrigation Benefit Program are also reducing dependency on ground water extraction.

- **Way Forward:**

▫ **Controlling over-extraction:**

- i. **Rationalizing Energy Pricing** - to make it expensive to withdraw groundwater
- ii. **Reforming MSP regime**
- iii. **Incentive Driven - Volumetric Water Pricing** - This will motivate the economy in water-use.
- iv. **Improving Water Use Efficiency in Agri-Sector:** Micro-irrigation, less water intensive crops.
- v. **Development of high yielding varieties** of crops such as pulses, millets etc. The increased productivity can play a crucial role in encouraging farmers to move away from water guzzlers.
- vi. **Enforcing Groundwater regulations** - To ensure groundwater is used in a sustainable manner.

▫ **Recharging Groundwater:**

- » **Protecting and rehabilitating wetlands; Watershed Management** (small and large check dams)
 - For e.g. the state of Maharashtra have started their own Managed Aquifer Recharge Program such as the Jalyukt Shivar.
- » **Rainwater Harvesting** in urban areas could be crucial in reducing the extraction of groundwater.
 - Effectively implementing Building Laws - to make it mandatory for large houses to provide water harvesting facilities.

▫ **Pollution Control:**

- » Reducing the Open Dumping of Municipal Solid Waste, controlling surface water pollution and strict action against water polluting industries.

- **Conclusion:**

- » Through these efforts, India can help to address the issue of groundwater depletion and ensure that this vital resource is available for future generations.

6. NATIONAL RIVER INTERLINKING PROJECT

- **Introduction**

- » The NRLP (previously called National Perspective Plan) is a large-scale civil engineering project that aims to link Indian Rivers by a network of reservoirs and canals and so reduce persistent floods in some parts and water shortage in other parts of India.

- **3 Components:** The inter-link project has been split into three parts: A Northern Himalayan River Interlink component; A Southern Peninsular Component; And starting 2005, an intrastate river linking component.

- **Managing Authority:** The project is managed by National Water Development Agency (NWDA), under the Ministry of Jalshakti.

- **Positives as claimed by proponents of River interlinking project**

- » **Improved Water Management:** In Monsoon type of climate, most of the rainwater is received within a few months and therefore its important to store this in reservoirs, canals etc, in order to use it for the whole year. This will improve irrigation and enhance food security; It would also reduce the vulnerability to draught.
- » **Electricity Generation (Hydro power)**
- » **Flood Control**
- » **Fisheries** - diversification of income generation

- » **Salinity and Pollution Control**
- » **Inland Navigation** - cheaper, less polluting, less accident prone, environment friendly means of transportation.
- **Key Concerns:**
 - » **Cost of Interlinking** (Land acquisition, construction etc.)
 - » **Ecological and Environmental Issues**
 - **Preventing all forms of flood may be harmful** as floods play an ecological role of recharging ground water, spreading alluvial soil, protecting biodiversity etc.
 - **Natural supply of nutrients, coastal ecosystem** etc might be affected.
 - **Reduction in sediment deposits** may negatively harm river deltas, which may increase coastal erosion and impact livelihood for a lot of people.
 - **Ensuring minimal ecological flow** in rivers is crucial for fighting river pollution, biodiversity conservation etc.
 - » **Displacement of people and economic harm to people dependent on aquatic ecosystem** dependent activities like fisheries, aquaculture etc.
 - » **Heavy Lifts** would be required due to topography of the peninsular river, which is going to have its own economic trade-offs.
 - » **International water sharing disputes may further be aggravated** (India, BD; India-China)
 - » **International Comparison**
 - i. Rhine-Main-Danube Canal
 - ii. Illinois Waterway
- **Conclusion:** The intentions of the project are right, but it's important that concerns raised by environmentalists and Supreme Court appointed Central empowered committee are taken into consideration to ensure minimum losses and maximum benefit out of the project

7. NATIONAL WATER POLICY, 2012

- **PYQ:** Enumerate the National Water Policy of India. Taking river Ganges as an example, discuss the strategies which may be adopted for river water pollution control and management. What are the legal provisions of management and handling of hazardous wastes in India? [100 words] (2013)
- **The Objective** of the National Water Policy, 2012:
 - » **Ensuring availability** of water for all human needs, including drinking, sanitation, and livelihood.
 - » Promoting **integrated water resource management** to optimize the water use and allocation across different sectors
 - » **Conserving** water resources through efficient and sustainable water use practices.
 - » Adopting a **participatory approach** involving local communities in water management decisions.
 - » Promoting **research and development** in the water sector to address emerging challenges.
- **Key Highlights:**
 - i. **Water as a Common Resource:** The policy recognizes water as a common resource to be held in public trust and advocates for its **equitable distribution**.
 - ii. **Integrated Water Resources Management (IWRM):** The policy emphasizes the adoption of IWRM, which involves managing water resources in a holistic manner, considering all aspects of water availability, demand, and quality.

- iii. **Basin-Level Planning:** The policy encourages the formulation of water management plans at the river basin level, recognizing that river basins are the natural units for water resource planning.
- iv. **Pricing of Water:** The policy promotes the concept of treating water as an economic good and advocates for the rational pricing of water to discourage wasteful use and promote conservation.
- v. **Rainwater Harvesting and Conservation:** To augment water availability, especially in water-stressed regions.
- vi. **Groundwater Management:** The policy highlights the importance of sustainable management of groundwater resources, given their critical role in meeting water demand, particularly in rural areas.
- vii. **Climate Change Adaptation:** The policy acknowledges the challenges posed by climate change on water resources and emphasizes the integration of climate change adaptation measures into water management strategies.

8. PRICING OF WATER IN INDIA: NEED OF OVERHAUL

- **Introduction** By 2050, the demand for water is expected to exceed the supply. In spite of this, there are inefficiencies in water usage in India. One of the key reasons for inefficient water use in India is the low cost or free availability of water.
- **About Water Pricing.**
 - » As per the **UN's Dublin Principle (1992)**, water is **an economic good and hence should reflect its scarcity value.**
 - » This has also been **recognised by successive National Water Policies** including the 1987, 2002 and 2012 policy.
 - » The **latest 2012 National Water Policy envisages** that **pricing of water** should reflect its efficient use and reward its conservation.
- **Current water pricing situation in India:** Constitutionally, water is a state subject.
 - » So far, there are less than 10 states/Uts which have not imposed any charges for irrigation water.
 - » The water charges imposed by a state depends on the kind of crop, area irrigated, number of times each crop is irrigated, and the total volume of water used. Similarly, the water charges for industries and the domestic sector vary widely.
- **Key problems in India's Water Pricing mechanism**
 - » **Absence of independent water regulating authorities** in all states except Maharashtra -> the determination of water charges is **Adhoc, non-consultative and non-transparent.**
 - » **Central government lacks framework for water pricing** - The **2016 Water Framework bill** had given some principles: Water for industrial, commercial and agricultural use -> **full economic pricing**; Water for domestic use -> **a graded pricing**; Water charges should be determined on the volumetric basis and taking into account the equity and efficiency.
 - » **Free/subsidized electricity** have allowed **overexploitation of ground water** in India's irrigation system.
 - » The **Ground Water Conservation Fee** is also not applicable on water utilization for agriculture.
- **Why moving to water pricing regime would be difficult?**
 - » **water shortage** has already impacted all the sectors -> charging for water may become politically unviable.
 - » **Entrenched political economy** - (removing MSP on rice may be politically disastrous)
 - » **Inherent design problems** as government doesn't control all the water supplies, especially for irrigation which consumes around 80% of India's water.
- **Way Forward**
 - » **Independent statutory water regulator** should be created in every state with mandate to regulate various water uses and pricing.
 - » Create a **framework for water pricing** -> quickly enact the 2016 water framework bill.
 - » The **Ground Water Conservation Fee notification** should be modified to cover irrigation sector as well.
 - » Separation of electric feeders for agricultural and non-agricultural purposes would be crucial for the same.
 - » **Center should incentivize** setting up of water regulator and levying of minimum water charges by states.

- » **Promote competitive and cooperative federalism** – CWMI should consider water pricing as a parameter.
 - » **Awareness generation** about significance of water pricing.
- **Conclusion:** Water pricing is a complex subject, and its imposition has huge political overtones. But it is the only long-term, sustainable solution to promote water efficient and equitable use this precious natural resource. The determination of water-use charges must be rational, consultative and transparent. And the role of the statutory water regulator should be recognised for the purpose.

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9. RIVER POLLUTION

- Current River Pollution Situation in India:

- » As per CPCB's **National Water Quality Monitoring Program**, the number of high priority (i.e. highly polluted) river stretches in the country has almost remained same **between 2018 and 2022**. This indicates that there is not much improvement in river water quality situation in the country

- Factors:

- » **Discharge of untreated or partially treated sewage and Industrial effluents** from cities/towns in their respective catchments is the main cause of river pollution in states.
- » **Illegal dumping of solid waste** on the banks of the rivers
- » **Shortage of STP/ETP Capacity:** As per CPCB report (March 2021), the sewage generation in urban areas is at 72,368 million liters/ day whereas total operational treatment capacity was only 26,869 MLD.
- » **Poor operations and maintenance of Sewage and Effluent Treatment plants**
- » **Non-points sources** of pollution
- » **Rapid Industrialization and Urbanization** is further compounding the issue.
- » **Min-Ecological flow** is not being ensured in many rivers.

- Key steps being taken

- » It is the responsibility of states/Uts/local bodies to ensure treatment of sewage and industrial effluents before it being discharged into water bodies.
- » **MoEF&CC** is contributing to conservation of rivers by **providing financial and technical assistance** for abatement of pollution in identified stretches of rivers in the country through the Central Sector Scheme of Namami Gange for rivers in Ganga Basin and the Centrally Sponsored Scheme of National River Conservation Plan (NRCP) for other rivers.
- » In addition, the government has recently released Detailed Project Reports (DPR) for the rejuvenation of 13 major rivers prepared by the **Indian Council for Forestry Research and Education (ICFRE), Dehradun** in consultation with the State Forest Departments and other line departments.
 - Work proposed under DPRs include afforestation of river banks, fighting soil erosion, recharging ground water, sequester carbon dioxide, catchment area treatment, ecological restoration etc.
- » Further, under MGNREGA, rejuvenation of small rivers is being prioritized.
- » In Addition, sewerage infrastructure is created under the AMRUT and Smart Cities Mission of MoHUA.
- » **Law and Regulations:**
 - As per the Environmental (Protection) Act, 1986 and the Water (Prevention and Control of Pollution), Act 1974, the industrial units are required to install effluent treatment plants (ETPs) and treat their effluents to comply with stipulated environmental standards before discharging into river and water bodies.
 - CPCBs, SPCBs and Pollution Control Committees (PCCs) monitor the industries with respect to treatment of effluent discharge standards and take action for non-compliance under the provision of various acts.

- **Way Forward:** See the topic of Namami Gange

10. NAMAMI GANGE

- Introduction

- » River Ganga is considered one of the holiest rivers of India. Ganga basin accounts for 47% of India's irrigated land and feeds around 500 million of population. Despite its importance, it is one of the **most polluted rivers** in the world.
- » There have been several initiatives to clean Ganga so far. **National Ganga Action Plan 1** was started in 1986, **NGA-2** in 1993 and later extended to other states. Till 2014, more than 4,000 crores had been spent. But the river had remained dirty.
- » So, when government launched the Namami Gange in mid-May 2015, there was a new hope.

- **Namami Gange Program** was launched from June 2014 to 31st March 2021 to rejuvenate River Ganga and its tributaries with a budget of Rs 20,000 Crores.

- » **In 2023**, Government approved Namami Gange Mission-II with a budgetary outlay of Rs 22,500 crore till 2026.

- **Key Features of Namami Gange**

- » It is the **biggest ever initiative** to clean Ganga with a Rs 20,000 crore, centrally funded, non-lapsable corpus.
- » **Eight Mains Pillars of Namami Gange Scheme**
 - » Sewage Treatment Infrastructure
 - » River Surface Cleaning
 - » Industrial Effluent Monitoring
 - » Ganga Gram
 - » Afforestation
 - » River Front Development
 - » Biodiversity Protection
 - » Public Awareness
- » **Improved Governance Structure under Namami Gange:**
 - » **Implementation by** NMCG and its state counterparts - State Program Management Groups (SPMGs).
 - » **National Ganga Council** (replaced NGRBA) which is headed by PM and has chief ministers of five ganga basin states - UK, UP, Bihar, Jharkhand and West Bengal.
- » **Synergy between different ministries** - Ministry of Jal Shakti have signed MoUs with 10 other ministries to synergize the activities under Namami Ganga.
- » **Focus on involvement of more stakeholders** including states, ULBs and PRIs, People and private sector (through PPP projects)
- » **4 Battalion of Ganga Eco-Task force** has also been envisaged to spread awareness and for protecting the river.
- » **Mains Focus** on Namami Gange is on **pollution abatement interventions** which include
 - » **Interception, diversion and treatment of waste water** through bio-remediation, in-situ treatment, innovation technologies, STPs, Effluent Treatment Plants etc.
 - » **Rehabilitation** of existing STPs
 - » Immediate short term measures for **arresting pollution at exit points on river front** to prevent inflow of sewage etc.

- **Other Steps under the Namami Gange Program**

- i. **Hariyali** is a plantation project along the stretch of Ganga in all five states through which it flows.
 - ii. **Ganga Gram Yojana**
 - To develop STP, toilets etc. in all villages along the river Ganga. Based on Sichawal model (a Punjab village) which is based on cooperation of villagers for water management and waste disposal.
 - Government will spend Rs 1 crore per village in this plan.
 - iii. **Smart Ganga Cities**
 - **Program for infra development along cities on Ganga river.**
 - iv. **Promotion of organic farming** in villages along the Ganga.
- **But as the recent CPCB report shows, Ganga River is far from being clean.** The **Key challenges** which are impacting the cleanliness of Ganga are:
- i. **Sewage Treatment has not reached its target** (only around 10% of the STP sanctioned under the initiative have been developed) -> Land acquisition issues; lack of upgradation of STPs, STPs not fulfilling quality standards; less focus on drainage system (thus large part of sewage is not reaching STPs).
 - ii. **Restoring the flow of river has remained a challenge:** Dams, overextraction for irrigation
 - iii. **Sludge control has not been successful** -> Open defecation has continued due to poor management of toilets built under Swachh Bharat Mission.
 - iv. **Cost overruns and under-utilization**
 - Delays and non-finalization of action plans in timely manner has led to **under-utilization** of government resources.
 - At the same time, Namami Gange projects are suffering from delays and thus cost overruns like previous program.
 - CAG has also come down heavily on centre for spending injudiciously on media blitzkrieg.
 - v. **Governance Glitches**
 - Though an elaborate governance mechanism has been created under the program, but it has remained inefficient.
 - For e.g. there are not a lot of details about how different ministries which signed MoUs with water resource ministry are contributing. NCG, Empowered Task Force and other monitoring bodies need to meet more regularly to ensure better implementation of the program.
 - vi. **Lack of Focus on distributaries** - lack of basin management approach has led to less focus on tributaries of Ganga.
- **Way forward:** The above challenges clearly indicate that our clean Ganga strategy needs an overhaul to be really effective. Some of the steps that can be taken include:
- » Fast tracking all the **STP construction** along with development of proper infrastructure to ensure sewage reaching the treatment plants.
 - » Govt also needs to take steps to **increase the flow of water** in the river.
 - Tweak hydroelectricity projects to consume less water; increase water use efficiency in agriculture etc.
 - » **Ending open defecation in letter and spirit** would require a post construction management scenario for toilets and increased focus on behavior change.
 - » An autonomous body for rejuvenation of Ganga is required. This body should have members from among experts well versed with rivers, rather than bureaucrats.

- » **Various monitoring bodies** like NCG, empowered task force etc. needs to **meet more regularly** to really make the project successful.
- » **Decentralization of program** will be crucial for improving the governance and hence making Namami Gange successful.
 - We need to strengthen ULBs and PRIs to deal with the river pollution at local levels.
- » **Every citizen** should understand that rivers belong to all of us and it is the joint responsibility of each and every individual to keep the river clean.
 - People who are living in the Ganga basin have to be involved to achieve the required results.
- **Conclusion**
 - » Cleaning of Ganga is an enormous task and should be seen as an ongoing process. It needs to be prioritized as it is not only a pollution and health issue. But, it will have various socio-economic advantages like improved agriculture, fisheries and other benefits.
 - » The clean-up of Thames in London and Rhine flowing through Europe suggest that cleaning Ganga is possible. The quality of water in Ganga will ultimately depend on the actions of millions of Indians who depend on it for nutrition, electricity and spiritual fulfillment.

11. UN WORLD WATER CONFERENCE, 2023

- **Practice Questions:**
 - » Critically Analyze the outcomes of UN World Water Conference, 2023 [10 marks, 250 words]
- The World Water Conference, 2023 was convened by the UN in March 2023. It was first such conference after 46 years.
 - » The last UN Water Conference was held in **1977**.
 - It resulted in first global 'Action Plan' recognizing that "all people", whatever their stage of development and social and economic conditions, have the right to have access to drinking water in quantities and of a quality equal to their basic needs.
 - This declaration led to several decades of global funding and concerted efforts to provide drinking water and sanitation for all. These actions substantially reduced the population without access to safe drinking water in much of the developing world.
- **Need of the World Water Conference:**
 - » **The 2023 Conference** was held in the following context of serious environmental issues - flooding, drought, a severity of climate change and a looming food crisis.
 - » It also marked a mid term review of the Water Action Decade 2018-2028 (to advance the water agenda by energizing existing programs and projects, and inspiring water actions to achieve the 2030 Agenda, in particular SDG 6, which envisages the sustainable management of water and sanitation for all.
- The conference was a watershed moment in the global water movement. It brought together 10,000 participants from all over the world to discuss the challenges and opportunities facing the water sector.
- **Targets with which conference was initiated:**
 - » To identify game-changing ideas;

- » To make recommendations to policymakers on how to speed- and scale-up change through capacity development, data and information, innovation, financing, and governance;
 - » To place water at the centre of the climate agenda in activities building up to the High Level Political Forum (HLPF) in New York in July and the COP28 climate talks to be held in Dubai later this year
- **What were the Conference Outcome?**
- » The Central outcome of the conference was the International Water Action Agenda, to which governments, multilateral institutions, businesses and NGOs submitted 700+ commitments to address water security issues. They included a \$50 billion commitment from Indian government to improve the rural water services under its Jal Jeevan Mission.
 - » **Some other Commitments announced at the event are:**
 - **Technology:** There were specific innovations in wastewater treatment or solar treatment of water in remote areas, and a number of proposals for incubation platforms, including the IBM Sustainability Accelerator, focused on water management.
 - **Data and Models:** Before every large investment, we must anticipate potential impact. Efforts like WMO's Hydrological Status and Outlook System, offered data analysis tools.
 - **Knowledge Sharing** – Solutions to water problems already exist, we need to spread this knowledge. One useful tool here is W12+ Blueprint, a UNESCO platform that hosts city profiles and case studies of programs, technologies, policies that addresses common water security challenges.
- **These commitments are** expected to inspire the collective political will, which is needed to address the many water challenges.
- Overall, the agenda is a comprehensive document that address the full range of water challenges from water scarcity and pollution to climate change and water security.
- **Criticisms:**
- » **No Binding Commitments** - Without binding commitments, expecting strong political will towards SDG goal 6 will be difficult.
 - » **Commitments are not ambitious enough** and **don't target the root cause** of the problem.
 - » **Finance Mobilization** will be a challenge: Meeting SDG-6 by 2030 will require **\$114 billion** per year.
 - » **Lack of clear targets and metrics** - it makes it difficult to track progress and hold governments accountable.
 - » **Lack of data about water capacity and utilization** would make the commitments difficult to implement.
 - There are 'serious limitations in our knowledge about the volume, flux and quality of water in lakes, rivers, soils and aquifers.
 - There are huge gaps in water usage data.
 - The metering of water has triggered resistance from India to Ireland because of concerns about equitable access and affordability of water services
- **Conclusion:**

- » If these challenges can be addressed, the Water Action, Agenda has the potential to make a real difference in the lives of millions of people around the world. The agenda provides a roadmap for how to address the water crisis, and it offers a clear vision for a water-secure future.

12. DEEP SEA MINING

- Introduction:

- » **Deep sea mining** is the process of retrieving mineral deposits from the deep seabed - the ocean below 200m.
- » It will be crucial to deal with increasing demand of metals powering energy transition (i.e. move to renewable energy).
 - Rare Earth Minerals and metals are fundamental to the renewable energy technologies driving the world's energy transition.
 - Copper or Nickel for batteries
 - Cobalt for electric cars
 - Manganese for steel production
 - Rare Earth Minerals for electronic components
- » While the demand for these metals are rising, resources are becoming scarcer.
- » As per International Energy Agency (IEA), if climate goals were properly pursued through the massive expansion of renewable energy, about five times as much lithium and four times as much cobalt would be needed by 2030.

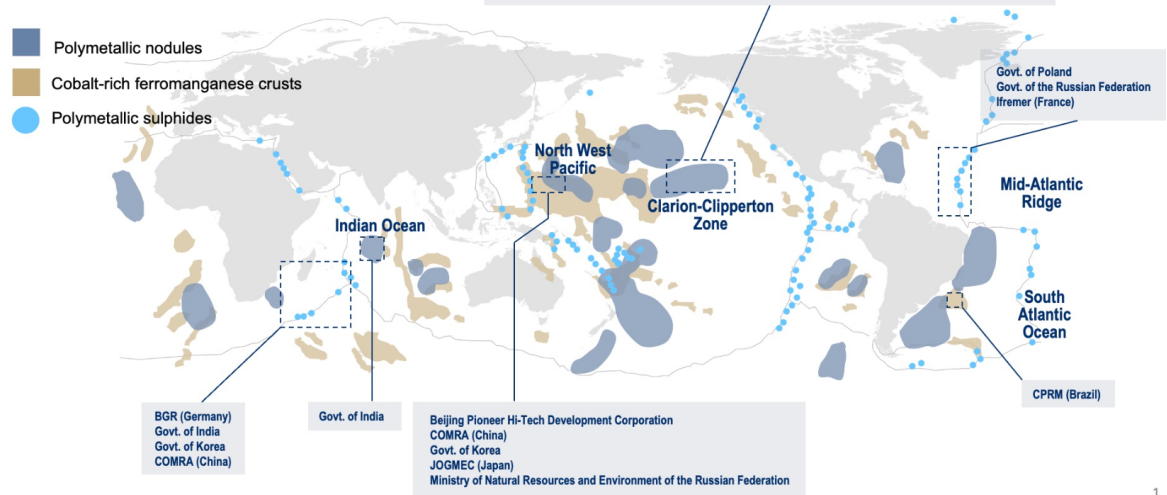
- Seabed holds valuable Manganese Nodules:

- » Manganese nodules (also called polymetallic nodules) are driving the rush to mine seabeds. These **potato-sized lumps** contain high proportion of Nickel, copper, manganese, rare earths and other valuable metals.
- » Although, manganese nodules are **not yet being mined anywhere in the world**, this could soon change as they practically lie directly on the seabed and can be easily extracted without breaking up rock layers or eroding the seabed.

- Potential Areas for mining

- » The best studied area is currently the seabed at between 3,500 and 5,500 meters in the **Clarion-Clipperton Zone** in the eastern Pacific Ocean near the US state of Hawaii.
 - Spanning thousands of kms, the areas contain more nickel, manganese and cobalt than any known area on land.
- » The basin in the Central Indian Ocean and the Seabed off the Cook Islands, Kiribati atolls, and French Polynesia in the South Pacific are also of interest for potential extraction.

Exploration for minerals in the Area



- Arguments supporting Deep Sea Mining:

- » **Resources**
- » It will be less damaging to the environment than extraction on land. It will emit 80% less GHG emissions.
- » It will support energy transition.
- » It will barely affect carbon reservoirs like forests and soils.
- » Since, deep sea mining will be mostly automated, it will avoid the exploitation of cobalt miners, including children, in Congo, when most of the world's cobalt is mined today.

- Criticism of Deep Sea Mining:

- » In mining, huge vacuum pumps will be used over the ocean floor to suck up the nodules - which will then be brought to the surface with a hose.
 - But, this will also destroy the living part of the seabed. All organisms, plants, bacterias etc. will be destroyed along with the nodules.
 - These organisms may also require manganese nodules to survive. These nodules can't be regenerated rapidly and would take millions of years for regeneration.
 - Clouds of sediments from the suction could cause enormous damage to ecosystems within a radius of several hundred kms.

- Who will authorize Deep Sea Mining?

- » **International Seabed Authority: ISBA**
 - In March 2023, ISBA have allowed companies to apply from July to mine the ocean floor.
 - Mining is expected to begin in 2025.

- Way Forward:

- » **Detailed Study of the Risk:** To understand the risk of mining comprehensively
- » **Impact Assessment:** Rigorous and transparent impact assessment based on comprehensive baseline studies
- » **Precautionary Principle** and **Polluter Pay Principle** should be implemented
- » **Transparent, accountable, inclusive and environment friendly** governance of Deep Sea mining

- **Conclusion:**

- » Precautionary principle will be crucial as deep-sea mining could irreparably harm ecosystems, and limit the many benefits the deep sea provides to humanity.

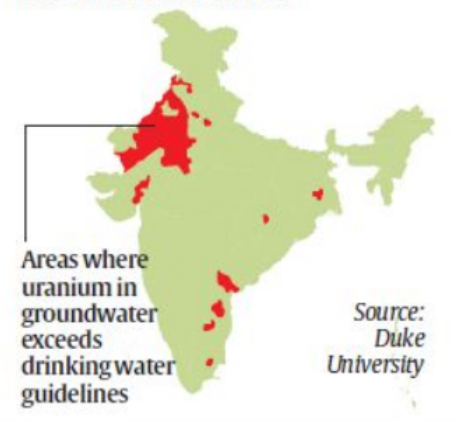
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13. SOME MISCELLANEOUS WATER POLLUTION TOPICS

1) URANIUM CONTAMINATION OF GROUND WATER

- **How much of Uranium in Water is acceptable?**
 - » **WHO** has set a provisional safe drinking water standard of 30 micrograms of Uranium per liter. This standard is also consistent with the US Environment Protection Agency Standards.
 - » **In India**, the **Indian Standard IS 10500: 2012** for Drinking Water specification has specified the maximum acceptable limits for radioactive residues as alpha and beta emitters, values in excess of which render the water not suitable. **But**, Individual radioactive elements have not been specifically identified.
- **Situation in India:** According to a study published in *Environmental Science and Technology* - **there is high Uranium Contamination in Ground Water of 16 Indian States**. **WHO** has also said that there is prevalence of concentration above 30 mg/lit of Uranium in some localized pockets of few states/UTs in the country.
- **Why the contamination?**
 - Ground Water Depletion and Nitrate Pollution may be aggravating the already present natural uranium contamination to dangerous levels.
 - **Process:**
 - Many of India's aquifers are composed of clay, silt and gravel carried down from Himalayan weathering by streams or uranium-rich granitic rocks. When over-pumping of these aquifers' groundwater occurs and their **water levels decline, it induces oxidation conditions** that, in turn, enhance uranium enrichment in the shallow groundwater that remains.
 - Though the **primary source is geogenic**, anthropogenic factors such as ground water table decline and nitrate pollution may further enhance uranium mobilization.
- **Impact**
 - Uranium contamination of drinking water may be responsible for chronic kidney diseases. Radioactivity is not an issue here, but the **toxicity** is.

RED IN RAJASTHAN



2) HEAVY METAL POLLUTION

- Heavy Metals and their Health Impacts

- » Heavy Metals are metals with relatively high densities, atomic weights, and atomic numbers.
- » Long term exposure to heavy metals may result in slowly progressing physical, muscular, and neurological degenerative process.
- » Once dispersed in the biosphere, these metals cannot be recovered or degraded. Hence, environmental effects of metal pollution tend to be permanent

- Sources of Heavy Metal Poisoning:

▫ Mining

- For e.g. mining releases chromium, cadmium, lead and mercury - all toxic heavy metals.
- Raniganj in West Bengal, Jharia in Bihar and Singrauli in Madhya Pradesh are considered some of the "hot spots" of metal pollution.

▫ Tailings

▫ Industrial Waste

▫ Agricultural runoffs

▫ Occupational exposures

▫ Paints

▫ Treated Timber

- Heavy Metal Situation of Indian Rivers:

- 75% of the river monitoring stations (across 117 rivers and tributaries and 688 stations) in India posted alarming levels of heavy toxic metals such as lead, iron, nickel, cadmium, arsenic, chromium, and copper.
 - **Ganga River** has high levels of lead, iron, nickel, cadmium, and arsenic. The report is an annual compendium of environment-development data and is derived from public sources.
 - Central Water Commission had tested water samples from 688 stations for heavy metals between Aug 2018 and Dec 2020.

- Way forward

- Control at Source
- Reduce use
- Proper disposal system
- Laws/rules -> strict implementation

3) ARSENIC POLLUTION

- Introduction

- » **Arsenic** is a odorless and tasteless metalloid which is widely distributed in the earth's crust.
- » **Arsenic contamination of the ground water is one of the most serious drinking water issue** being faced in India.
- » **BIS** stimulates a permissible limit of 0.01 mg/L of arsenic in water. But, as per the latest CGWB study, **21 states** across the country have pockets of arsenic levels higher than this limit.

- Key Areas impacted by Arsenic Pollution in India

- » The states in Ganga-Brahmaputra-Meghna river basin are the most affected. They include - UP, Bihar, Jharkhand, WB, and Assam.
- » Other arsenic affected areas include Punjab, Haryana, Manipur, Chhattisgarh and Karnataka.

- Sources of Arsenic Pollution

- » Arsenic is introduced in soil and groundwater through weathering of rocks and minerals followed by subsequent leaching and runoff.
- » **Anthropogenic sources** - coal fired power plants, burning vegetation, and Volcanism.
- » **Ground water contaminated with Arsenic** is also entering food chain.
 - The chemical has found its way into rice, wheat and potato. An unique observation was that in several samples, arsenic content in food items was higher than that in drinking water.

- Impact

- » Long-term intake of arsenic polluted water leads to arsenic poisoning or arsenicosis, with cancer of skin, bladder, kidney or lung or diseases of skin, blood vessels of legs and feet.

• Key Recent steps:

- Under Jal Jivan Mission (Har Ghar Nal se Jal), since, planning, implementation, and commissioning of piped water supply scheme based on a safe water source may take time, purely as an interim measure, state and Uts have been advised to install **community water purification plants (CWPP)** especially in Arsenic and Fluoride affected habitations to provide potable water to every household at the rate of 8-10 litres per capita per day to meet their drinking and cooking requirements.

• Way forward

- **Water treatment** -> advanced methods
- **Reducing contamination** through anthropogenic sources.
- **Arsenic Mitigation strategies:** Using non-contaminated or less contaminated well, **appropriate treatment** to reduce arsenic content in drinking water.
- **Adopting Rainwater harvesting/watershed management** etc. would reduce water stress but also ensure arsenic free supply of water (naturally or through filtration).
- **More comprehensive mitigation approach** - also check arsenic used for agricultural produce.
- **More R&D** on arsenic contamination of our food chain and its health impacts.

4) RADIOACTIVE POLLUTION IN WATER

- Details

- » Radioactive pollution of water is a newly emerging, but grave concern of water pollution and human health.
- » Radioactive elements are naturally found in earth's crust. Percolation of naturally occurring radioactive materials (NORM) from the soil sediments to the aquifer causes groundwater contamination.
- » **Anthropogenic sources include:**
 - **Nuclear weapon investigation;**
 - **nuclear calamities;**
 - **nuclear power house;**
 - **dumping of radioactive waste are the major sources.**
 - **Use of radioisotopes in industries and scientific laboratories are the minor sources.**
- » This pollution is more prevalent in groundwater as compared to surface water since it is much exposed to radioactive elements found in rocks. Sometimes magma also releases radioactive gases in environment.

- **A small amount** of radiation is found in all types of water but the extended amount of radiation is harmful to human health.

- Harmful Impacts of nuclear radiation:

- » Immediate: recoverable consequences distressing skin, lungs, genitals, and causing of hair fall.
- » Long standing: permanent outcomes such as various infections like radiation damage, bone marrow fatality, cataract initiation, cancer stimulation, cholera, etc.
- » Genetic effects: ionizing radiation induces mutations in germ cells

- WHO guidelines:

- » Guidelines for drinking water quality and a permissible limit of reference dose level of 0.1 micro-sieverts per year.

- Way Forward:

- » **Safe Water Supply:** Proper Analysis and monitoring of radioactive pollutants
- » **Prevention and precaution measures** can check the anthropogenic sources of radioactive contamination in water.
- » **Various treatment methods** like aeration, reverse osmosis, ion exchange and granule carbon absorption are effective remedial measures for treating the radioactive contaminated water.
- » **International Cooperation:** With increasing exposure to radioactive pollution all over the world, it's important that the international community collaborates together through an International Convention to control it.

5) FUTURE POLLUTION TOPICS – LEAD POISONING, MERCURY POISONING, MARINE POLLUTION, HIGH SEAS TREATY ETC.

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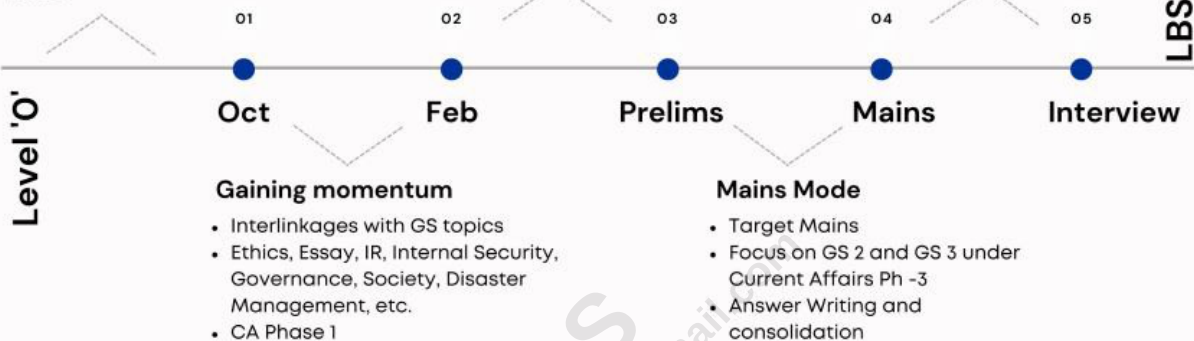
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